UNIT-I: Introduction to Artificial Intelligence

Part A: Introduction to AI

Q1. Define Artificial Intelligence. Explain the characteristics of an intelligent system and discuss the objectives of AI.

Q2. Explain different categories of AI based on capability and functionality with suitable examples.

Part B: AI Problems

Q3. Differentiate between conventional programming problems and AI problems using appropriate examples.

Q4. Explain the following AI problems with suitable applications(any two)

- Game Playing
- Natural Language Processing
- Computer Vision
- Robotics
- Expert Systems

Part C: Intelligent Agents

Q5. Define an intelligent agent. Explain the components of an agent with the help of a neat block diagram.

Q6. What is PEAS? Explain the PEAS description for (any two) of the following:

- Self-driving car
- Medical diagnosis system
- Vacuum cleaner
- Chess-playing agent

Q7. Explain different types of intelligent agents

- Simple Reflex Agent
- Model-Based Reflex Agent
- Goal-Based Agent
- Utility-Based Agent
- Learning Agent

Part D: Rational & Problem solving Agents

Q8. Define a rational agent. Explain the concept of rationality with diagram and with suitable examples.

Q9. Explain the following components of a problem formulation with an example :

- Initial State
- State Space
- Actions
- Transition Model
- Goal Test
- Path Cost

Part E: Nature of Environments

Q10. Classify environments based on the following characteristics:

- Fully Observable vs Partially Observable
- Deterministic vs Stochastic
- Episodic vs Sequential
- Static vs Dynamic
- Discrete vs Continuous
- Single Agent vs Multi-Agent

Part F: AI / ML / DL

Q11. Compare Artificial Intelligence, Machine Learning and Deep Learning.

Case Study 1

Q12. A smart vacuum cleaning robot is designed to clean rooms automatically. It detects dust using sensors and moves accordingly.

Identify and explain:

- Agent
- Environment
- Sensors
- Actuators
- Performance Measure
- Type of Environment

Case Study 2

Q13. A self-driving car detects pedestrians, traffic signals and nearby vehicles using cameras and sensors.

Explain:

- PEAS description
- Nature of environment
- Type of intelligent agent
- Why it is considered a rational agent?

Case Study 3

Q14. An AI chatbot answers customer queries in an online shopping portal.

Discuss:

- Agent type
- Sensors and actuators
- Performance measure
- Learning capability
- AI techniques involved

Case Study 4

Q15. A hospital develops an AI system for disease diagnosis using patient symptoms and medical history.

Explain:

- Why it is considered an AI system
- Problem formulation
- Agent architecture
- AI vs Machine Learning in this application

Case Study 5

Q16. A delivery drone has to reach a destination while avoiding obstacles.

Formulate this as an AI search problem by identifying:

- Initial state
- Goal state

- State space
- Actions
- Path cost
- Environment type

Case Study 6

Q17. An autonomous agricultural robot identifies weeds and sprays pesticides only where required. Discuss:

- Type of AI application
- Agent structure
- Rationality
- Benefits to society